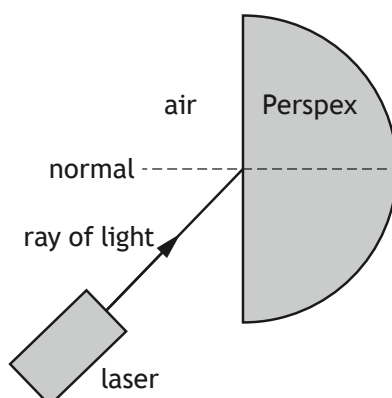


12. A student is investigating refraction of light using the apparatus shown.



- (a) The student measures a number of different angles of incidence θ_i and the corresponding angles of refraction θ_r .
Suggest how the student's measurements should be processed to find a reliable value for the refractive index of Perspex for the light from the laser.

2

- (b) The student determines the refractive index of the Perspex block for this light to be 1.50.

Calculate the critical angle of Perspex for this light.

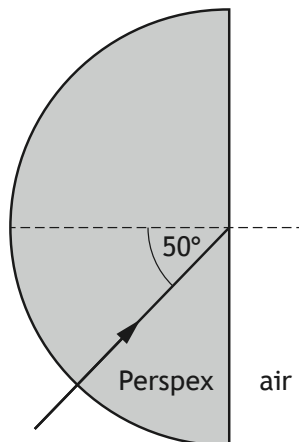
3

Space for working and answer



12. (continued)

(c) The student now directs the ray of light into the Perspex block as shown.



Complete the diagram to show the path of the ray of light after it is incident on this Perspex-air boundary.

Mark on the diagram the values of any relevant angles.

(An additional diagram, if required, can be found on *page 53*.)

2

[Turn over

